Bridge Day 4 INSTRUCTOR KEY

Comparisons, Exact Boundaries, and Ordered Branches

Learning sequence: Predict - Trace - Explain - Test - Interpret

Thursday, July 9, 2026 • 20 points • target 16 minutes

Name		**UIN**		**Section**	

Assessment boundary and evidence source

Contract: 07a04826c975
Cumulative foundations: input conversion, comparisons, inclusive boundaries, if/elif/else, f-strings
Scaffold level: complete ordered four-case classifier
Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.15, 18.16, 18.17.
Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Evaluate exact-boundary comparisons. - 5 points

```
low = 10
high = 15
value = 15
within = low <= value <= high
above = value > high
print(within, above)
```

Question: Evaluate every comparison and explain which operator controls the upper boundary.

Model response: 10 <= 15 and 15 <= 15 are both True, so within is True. The strict comparison 15 > 15 is False. Exact output: True False. The <= operator includes equality at high.

Item 2. Trace the first matching ordered branch. - 5 points

```
reading = 25

if reading > 25:
    status = "high"
elif reading >= 20:
    status = "normal"
else:
    status = "low"

print(status)
```

Question: Name the first matching branch, give exact output, and state what happens at 25 and 25.1.

Model response: At 25, reading > 25 is False and reading >= 20 is True, so the elif branch assigns normal. At 25.1, the first branch assigns high. Exact supplied output: normal.

Complete program - 10 points

- Read temperature as a float.
- Print invalid when temperature is below -40 or above 80.
- For valid input, classify cold below 15, warm from 15 through 30 inclusive, and hot above 30.
- Print exactly Status: followed by the classification using one if/elif/elif/else chain.

Program workspace

Model program

```
temperature = float(input())

if temperature < -40 or temperature > 80:
    print("invalid")
elif temperature < 15:
    print("Status:  cold")
elif temperature <= 30:
    print("Status:  warm")
else:
    print("Status:  hot")
```

Test input, expected result, and why this test matters

Reasoning and test evidence The invalid branch protects the announced domain first. Once input is valid, the earlier cold branch removes values below 15, so temperature <= 30 correctly represents the remaining inclusive warm interval. Boundary evidence: -40 is cold, 15 and 30 are warm, 30.1 is hot, and 80.1 is invalid.
Scoring guidance: 2 input, 2 invalid domain, 3 ordered boundaries, 2 mutually exclusive structure/exact output, 1 boundary evidence.

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.